

Confirmatory forecasting verdict

The frozen histogram gradient boosting model was evaluated once on the untouched confirmatory period. It reduced one-step-ahead mean absolute error relative to persistence in aggregate, during the frozen peak state, and in every prespecified temporal block.

27.6%

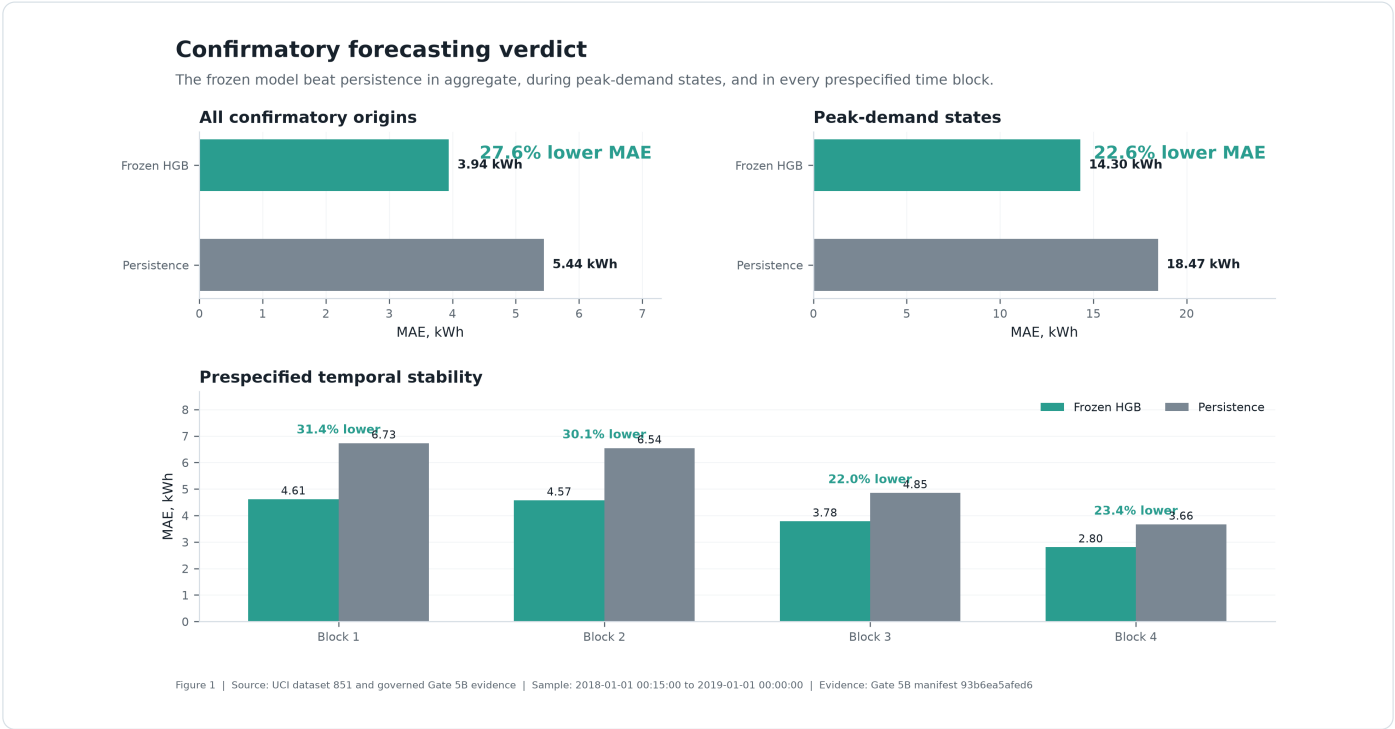
AGGREGATE MAE REDUCTION

22.6%

PEAK-STATE MAE REDUCTION

4/4

POSITIVE TEMPORAL BLOCKS



CONFIRMATORY DECISION

The frozen model reduced one-step-ahead energy-demand MAE by 27.6% relative to persistence, retained a 22.6% advantage during peak-demand states, and outperformed the reference in every prespecified temporal block.

Candidate MAE: 3.94 kWh | Persistence MAE: 5.44 kWh | Confirmatory origins: 7,004

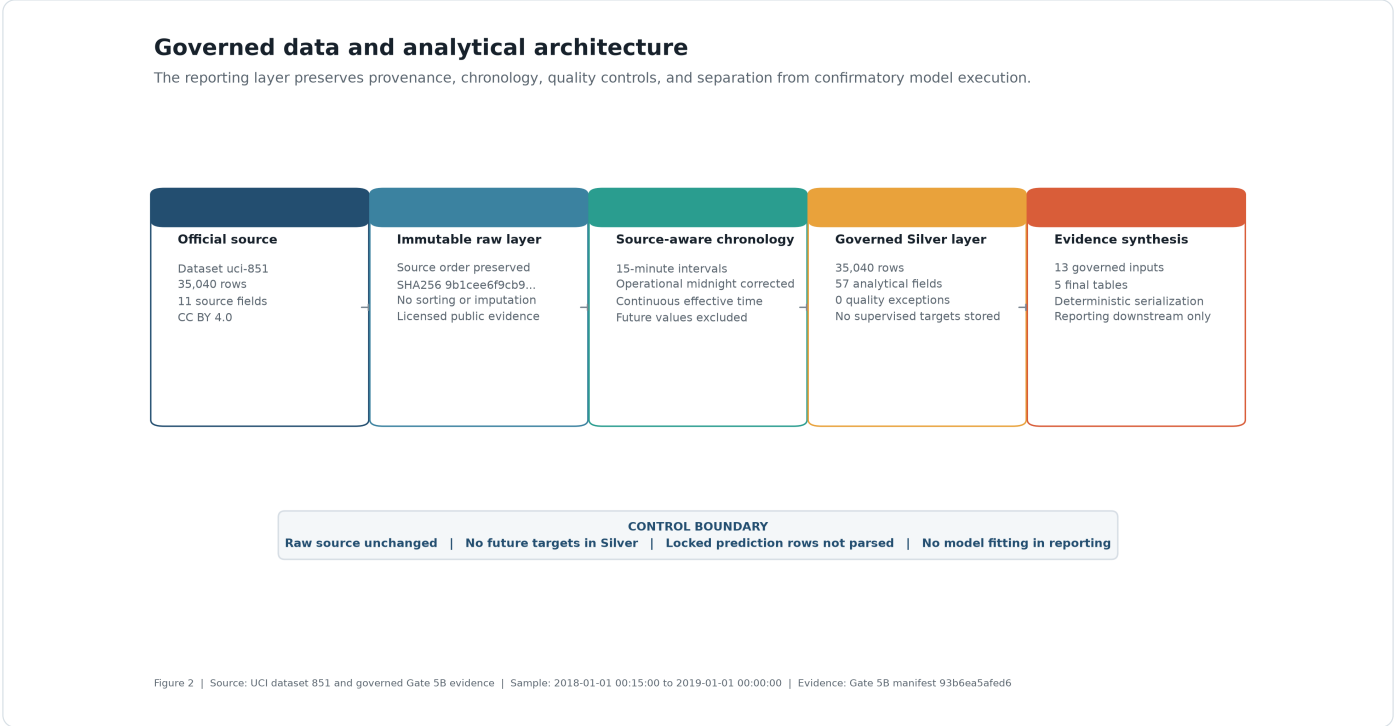
Governed data and analytical architecture

The governed UCI-851 source contains 35,040 licensed 15-minute observations. Source order, timestamps, analytical transformations, quality controls, and evidence identities remain explicit throughout the pipeline.

35,040
LICENSED OBSERVATIONS

15 min
SAMPLING INTERVAL

0
AGGREGATE QUALITY FLAGS



Model ladder and chronological validation

Model selection used four expanding-window chronological folds and 7,004 validation origins. Persistence remained the formal reference, and the locked confirmatory period was excluded from selection, tuning, and re-estimation.

4

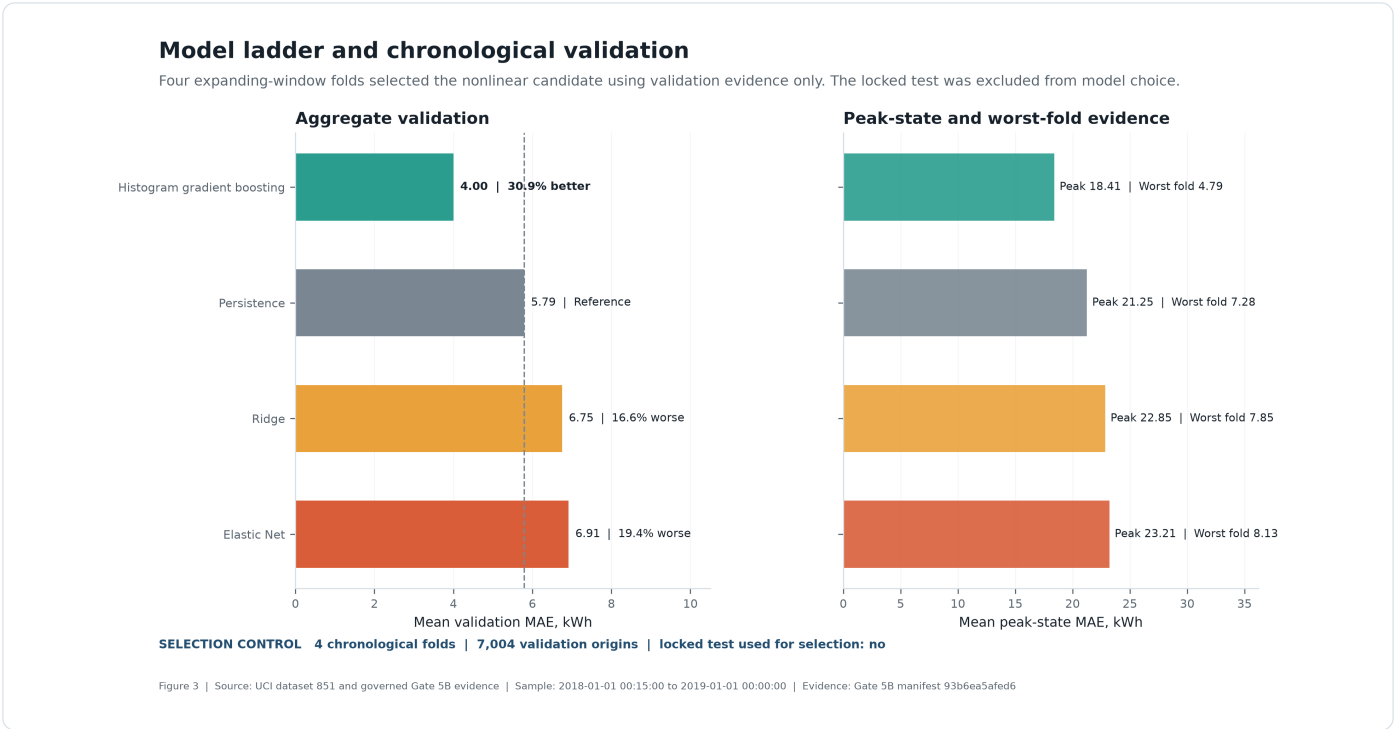
EXPANDING-WINDOW FOLDS

7,004

VALIDATION ORIGINS

Validation only

SELECTION BASIS



Locked-test stability and peak-state robustness

The single authorized confirmatory execution covered 7,004 origins. Peak-state evidence contained 761 rows above the frozen 80.96 kWh threshold, while relative MAE improvement remained positive across all four equal temporal blocks.

1

AUTHORIZED EXECUTION

7,004

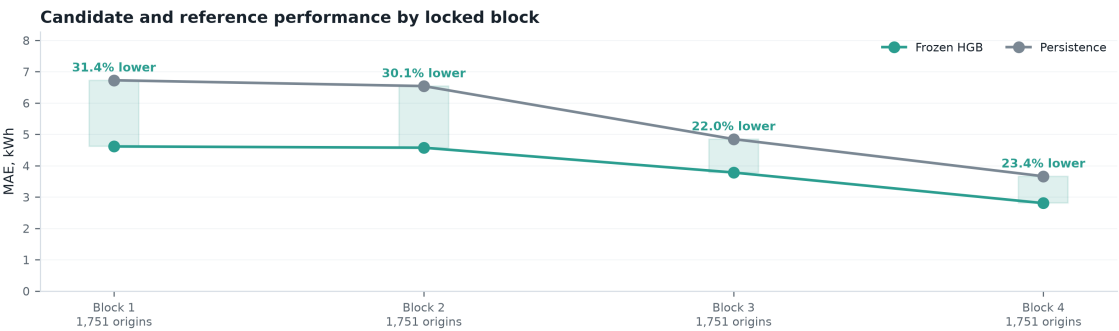
CONFIRMATORY ORIGINS

761

PEAK-STATE ROWS

Locked-test temporal stability

The frozen model retained lower MAE in all four equal, prespecified blocks and during the governed peak state.



Confirmatory summary
27.6%
aggregate MAE reduction
7,004 confirmatory origins

22.6%
peak-state MAE reduction
761 governed rows

Locked evaluation boundary
Evaluation origins: 28,032 to 35,035
Four equal blocks: 1,751 origins each
Peak threshold: 80.96 kWh
Evaluation count: one

Figure 4 | Source: UCI dataset 851 and governed Gate 5B evidence | Sample: 2018-01-01 00:15:00 to 2019-01-01 00:00:00 | Evidence: Gate 5B manifest 93b6ea5afed6

TEMPORAL STABILITY

Relative MAE improvement remained positive in every prespecified block, ranging from 22.0% to 31.4%.

PEAK-STATE BOUNDARY

The frozen 761-row peak subset retained a 22.6% advantage over persistence. The threshold was not changed after confirmatory execution.

Evidence governance and model boundaries

Confirmatory evidence is closed and append-only. The single-use authorization was consumed, no re-estimation occurred, and the retired evaluator is neither imported nor invoked by the reporting layer.

Consumed

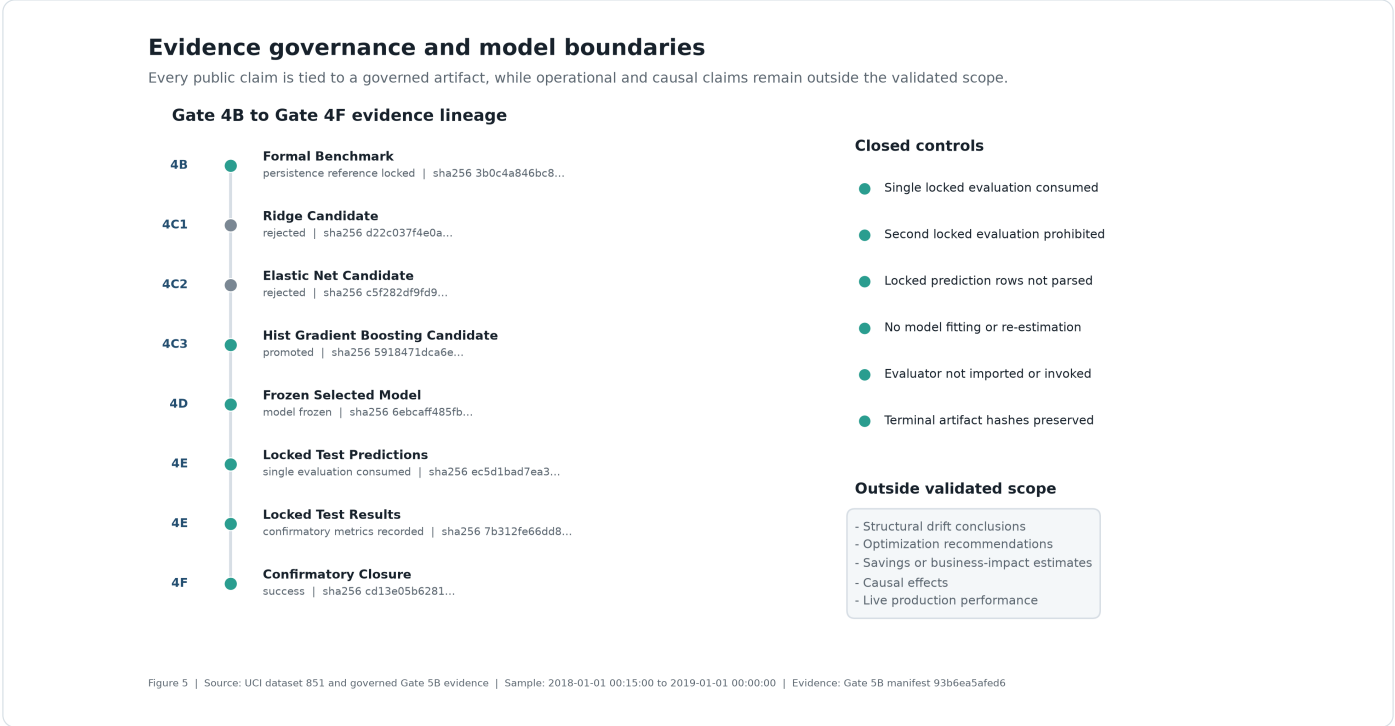
SINGLE-USE AUTHORIZATION

No

RE-ESTIMATION PERFORMED

Retired

EVALUATOR STATUS



VALIDATED SCOPE

- One-step-ahead energy-demand forecasting.
- Governed UCI sample and frozen chronology.
- Frozen feature and target contracts.
- Closed terminal confirmatory evidence.

OUTSIDE VALIDATED SCOPE

- Structural drift conclusions.
- Operational optimization recommendations.
- Savings or economic-impact estimates.
- Causal or live-deployment performance claims.

EXEC 9d47bd598b08... | PRED ec5d1bad7ea3... | RESULT 7b312fe66dd8...